Task 10 - Theory of chaos: Logistic map, chaotic numbers and their prediction

Create a bifurcation diagram that uses the logistic map function, which is defined as follows:

$$x_{n+1} = a \cdot x_n \cdot (1 - x_n)$$

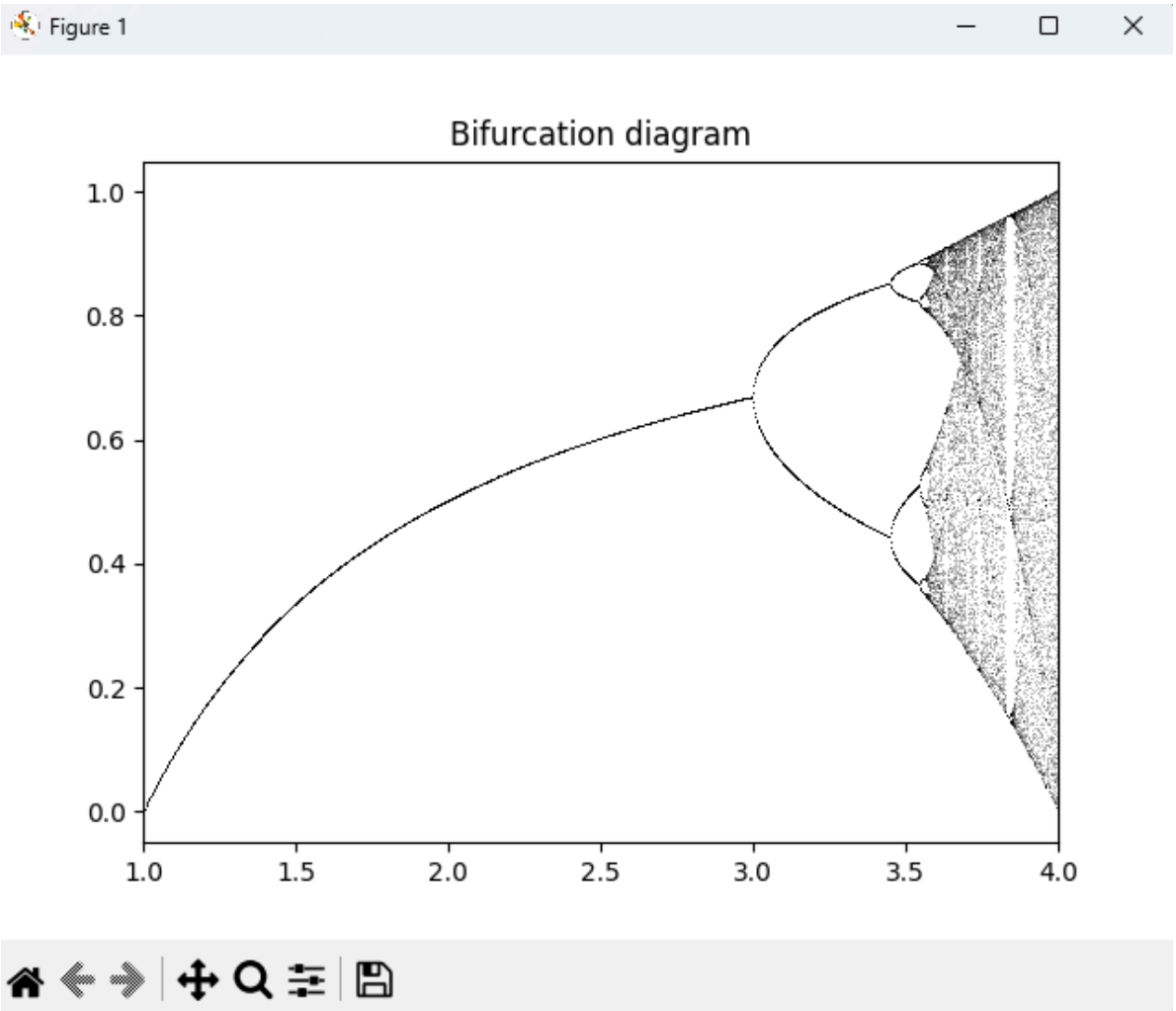
You can generate the bifurcation by using the number of samples, iterations and number of transitions. As a start of a values you can use linspace from numpy in python:

a_values = np.linspace(0, 4.0, 1000)

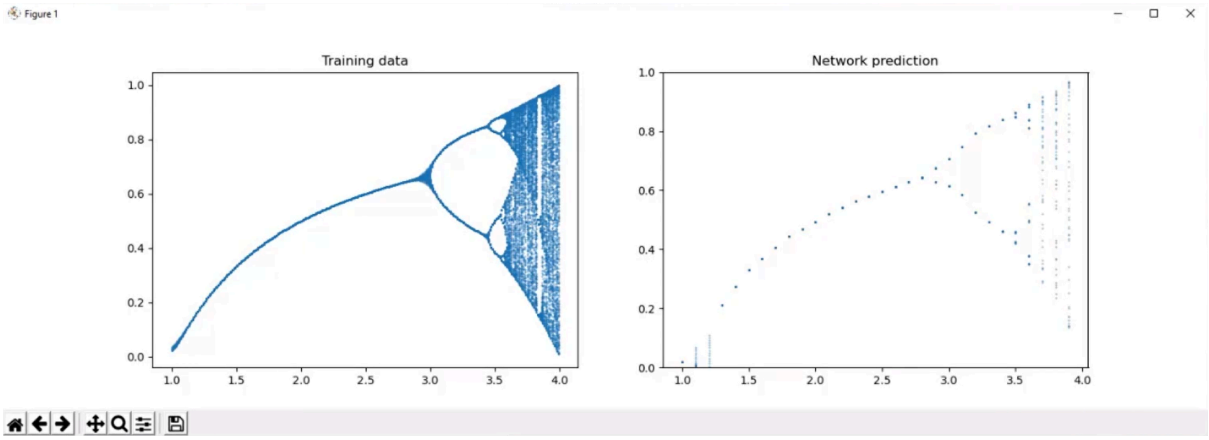
Instructions

1. Implement the logistic map
2. Visualize the bifurcation diagram for different values of the parameter a
3. Use neural network to predict the number of the logistic map
4. Visualize the bifurcation map predicted by the neural network

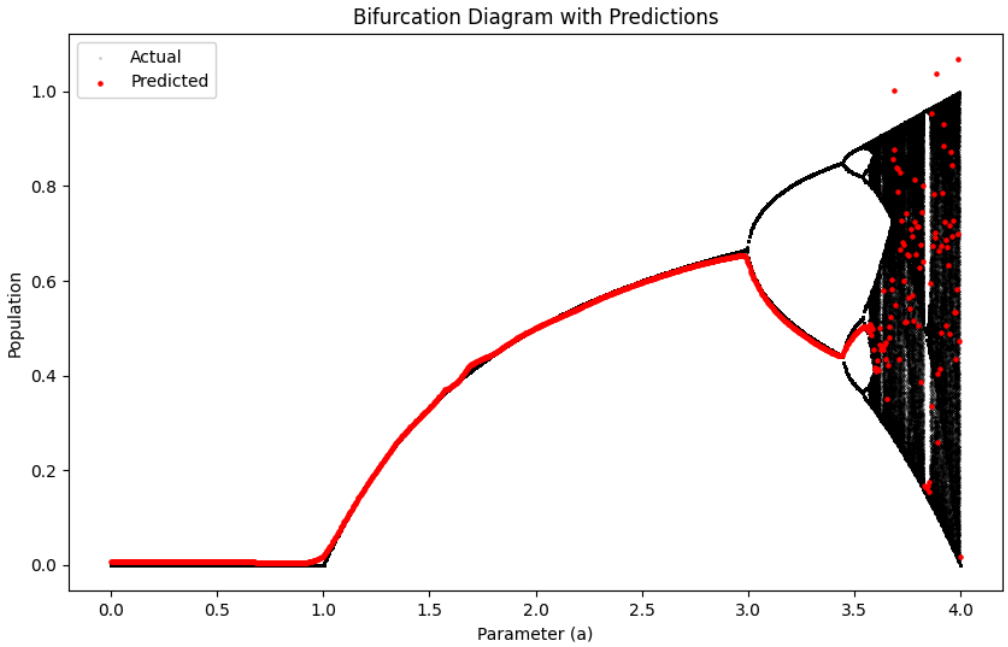
If you won't be able to create a full prediction you can just make some prediction like is on image Example 3. As an exception you can use any package for the prediction by neural network, so keras, pytorch or anything else can be used.



Example 1: Bifurcation diagram



Example 2: Training data on the left and network prediction on the right



Example 3: Prediction point by point on a bifurcation diagram of the logistic map